A brief Introduction To GIS



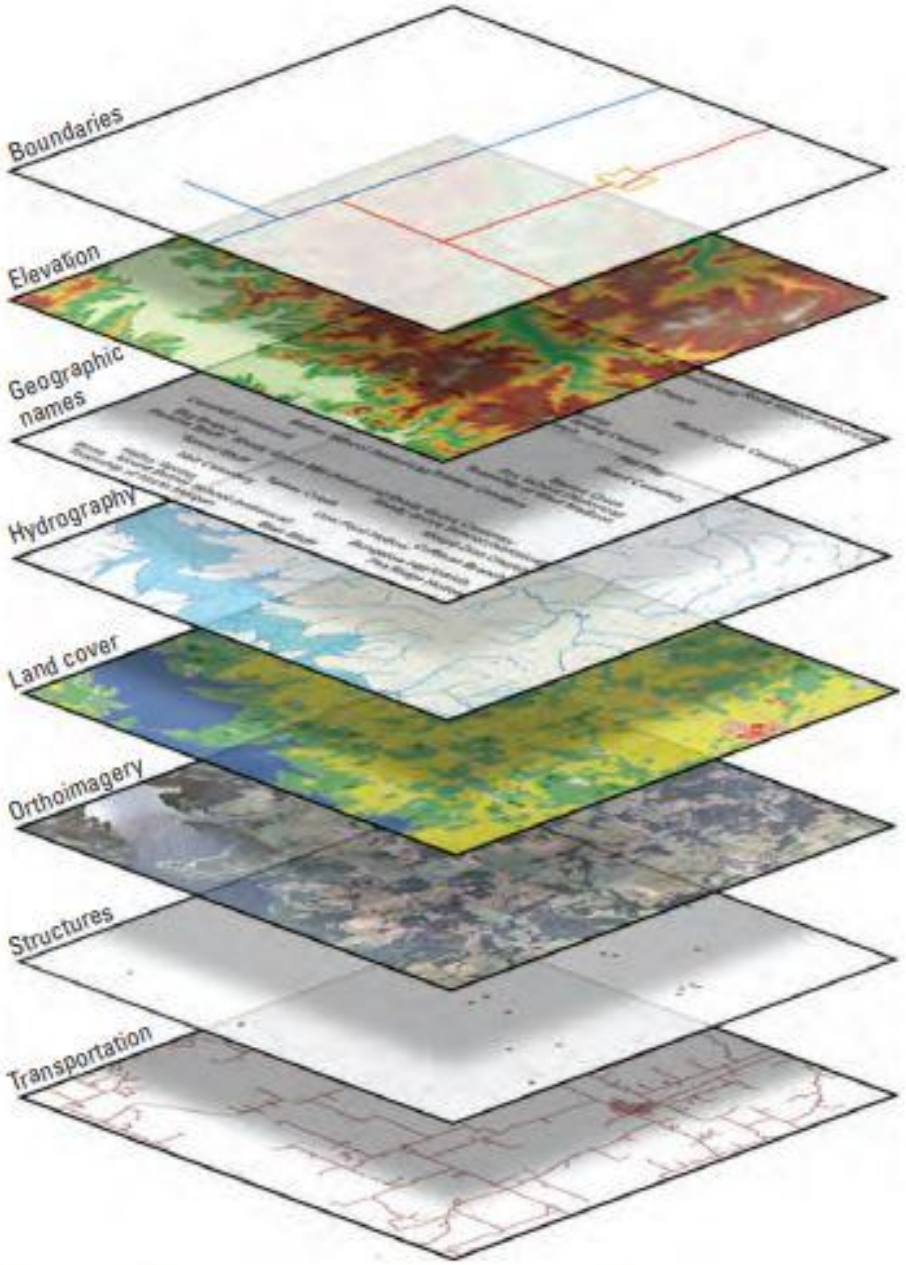


Figure 1. Eight base layers of *The National Map*.

What is GIS?
Geographic Information System

store, manage, analyze, edit, output, and visualize geographic data in a geospatial database

In a GIS, you connect ***data*** with ***geography***

-GISgeography.com

Two popular GIS programs



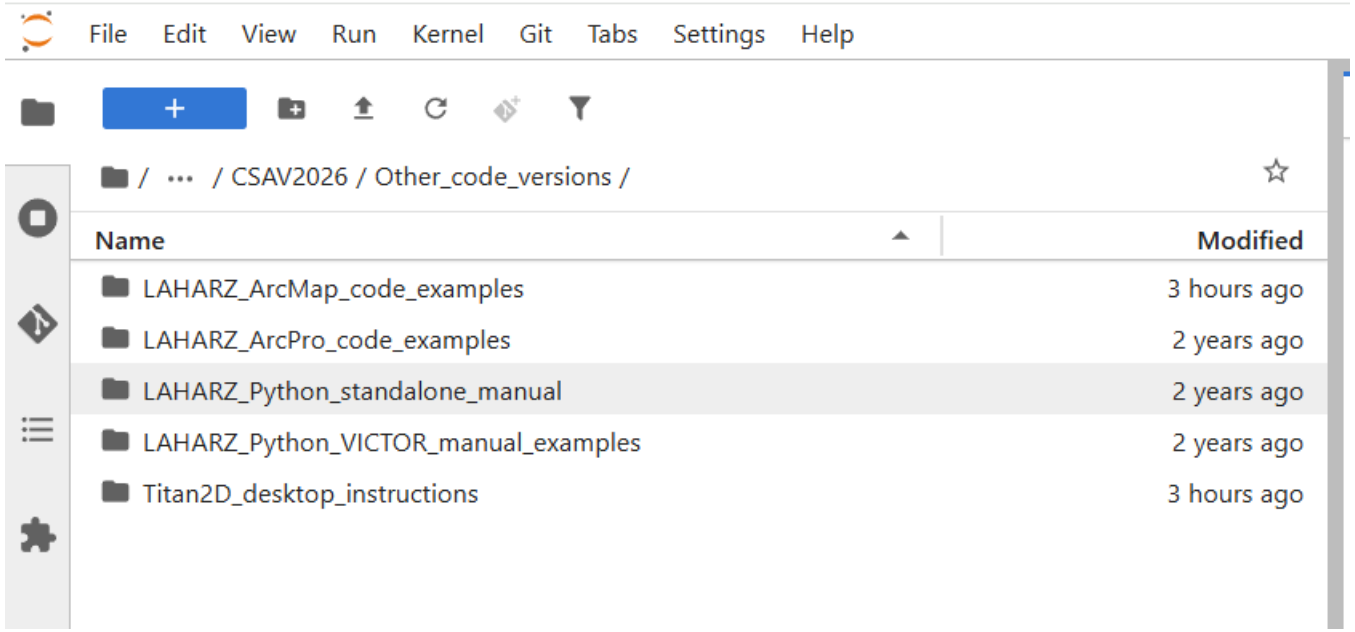
- Expensive, commercial
- Windows (but there are work-arounds)
- More stable, consistent, compatible; less flexible, less customizable
- Cannot open some QGIS files
- Until 2024, it was the ONLY way to run LAHARZ!



- Free, open source
- Windows, Mac, Linux, Android
- More flexible, customizable (plug-ins, scripting), but can have bugs
- Can open ArcGIS formatted files
- Thanks to Keith Blair, there is now an open-source, stand-alone python version of LAHARZ!

Two popular GIS programs

- For this year's course, we will use QGIS and open-source versions on models on the VICTOR platform
- However, the ArcGIS course materials (ArcMap and ArcPro) and LAHARZ code are still available and will be provided to you in the class files
- All model codes are also available as stand-alone version you can install



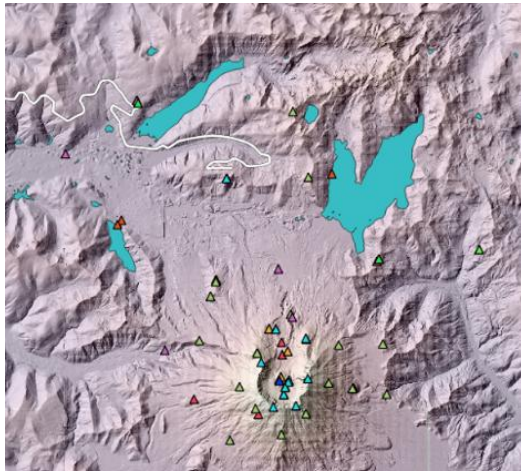
Representing Geographic Features

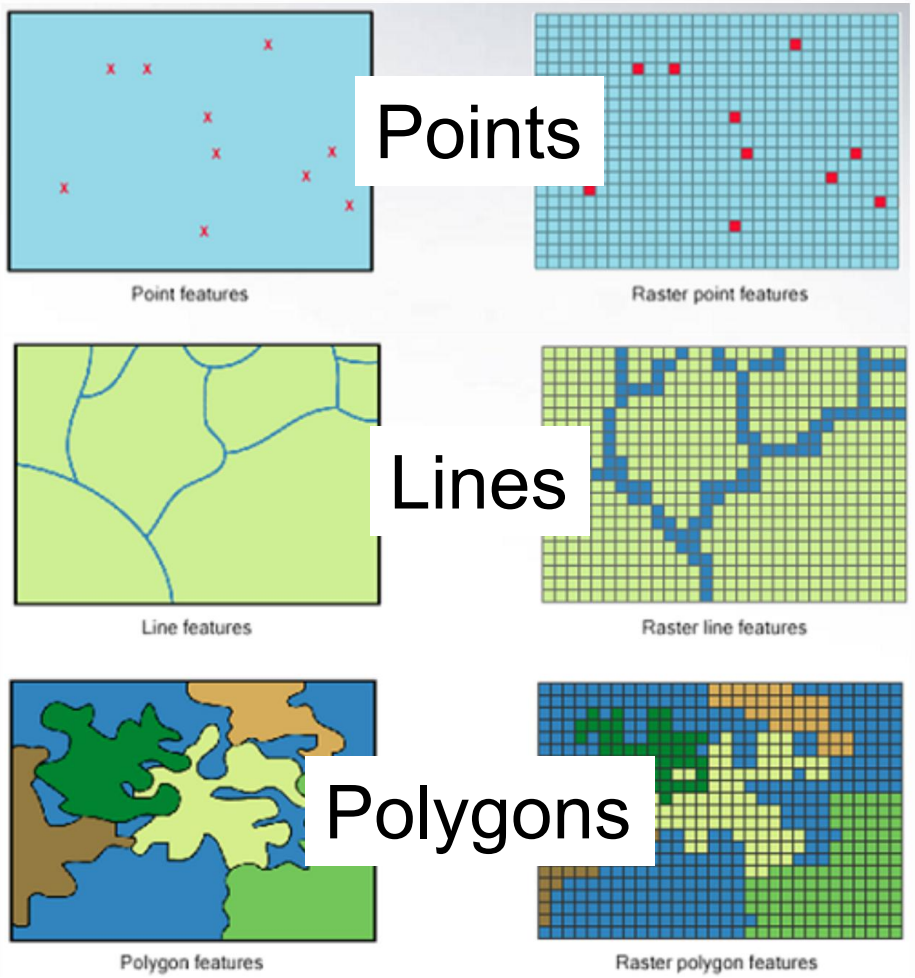
Vector Data

Raster Data

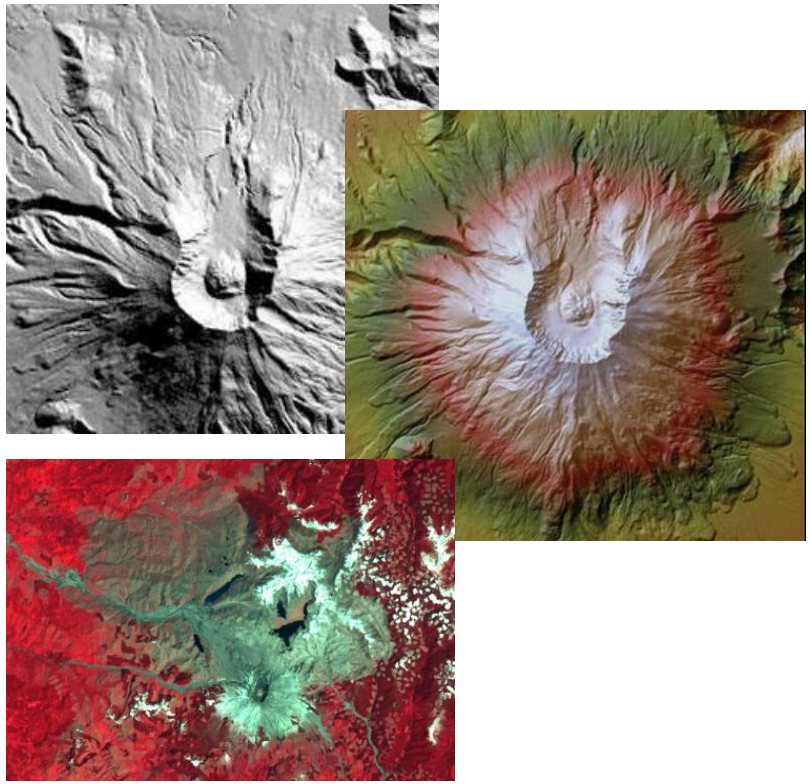
"Raster is faster; Vector is correct-er"

Features like road networks, lakes, park boundaries, station locations, etc. are usually vectors (in ArcGIS these are shapefiles)





Digital elevation models (DEMs), hillshades, satellite images are all rasters



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GIS works when you combine geographic location with attribute information

Where is it?

Spatial Coordinates
(204654, 3984988)

What is it?

Attributes

FID	Shape *	Volcano	Instrument	Solar?
024	point	Kilauea	Seismometer	Yes

and you display it with scale, direction, and system of measurement

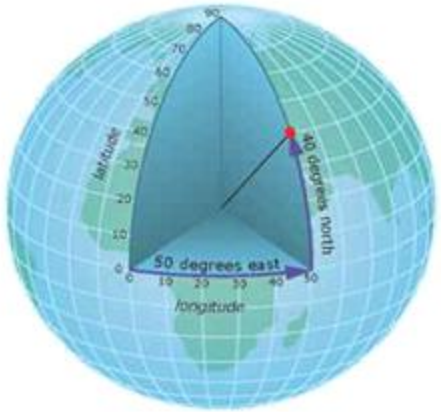
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Overview: Coordinate system types

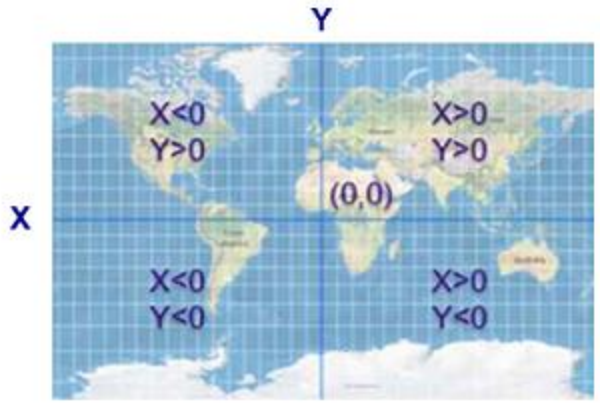
Geographic Coordinate Systems (GCS)

- Defines ***where*** *the data is located on the earth*
  - Angular measurements
  - Latitude & Longitude
  - Major variable: **datum**



Projected Coordinate Systems (PCS)

- Tells the data ***how*** *to draw on a flat surface*
  - Linear measurements
  - Feet & Meters
  - Major variable: **flattening method**



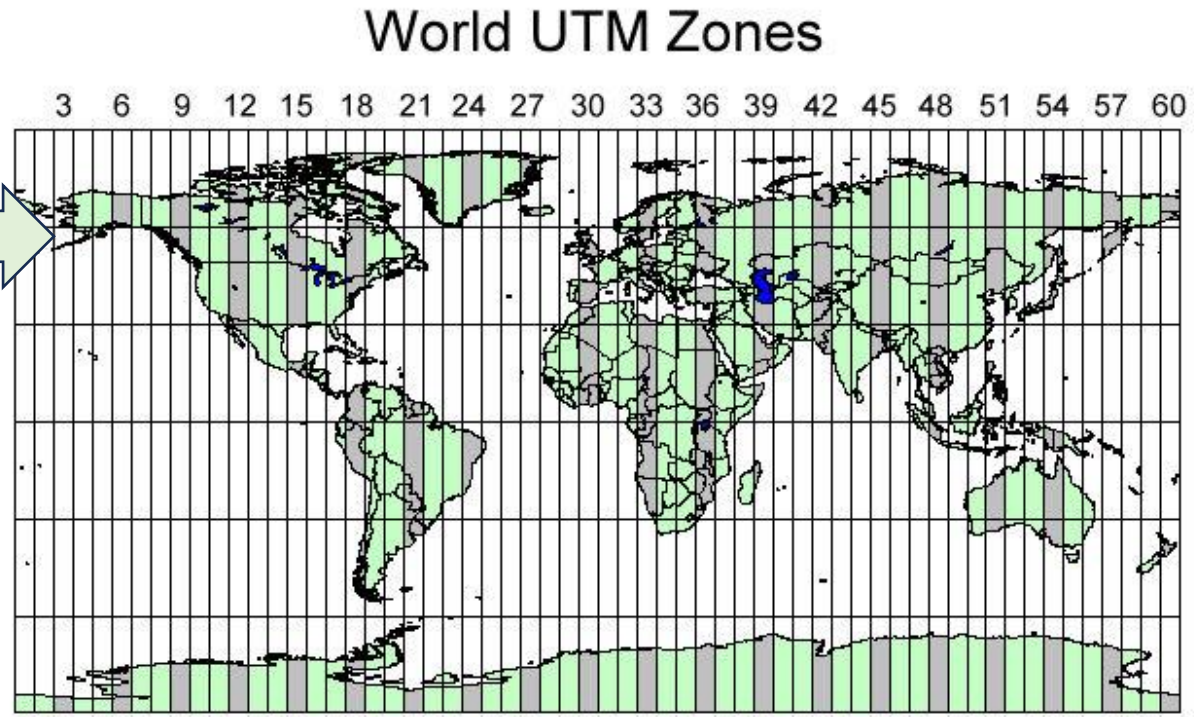
Projections can preserve size, shape, and distance, but not all three at the same time!

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- For the ArcMap and ArcPro versions of LAHARZ, the DEM needs to be in UTM
- For the VICTOR and stand-alone python versions, projections are handled internally so you can use the GCS DEM
- Titan2D needs a UTM DEM
- For your projects, you will need to find the UTM zone for your target volcano

Universal Transverse Mercator (**UTM**)

- Divides earth into 60 zones and projects each to the plane as a basis for its coordinates
- Specifying a location means specifying the zone and the *x*, *y* coordinate in that plane.
- Uses a grid-based, cartesian system so calculating distances is easy (which is why we use it for modeling)



DEMs on VICTOR



OpenTopography

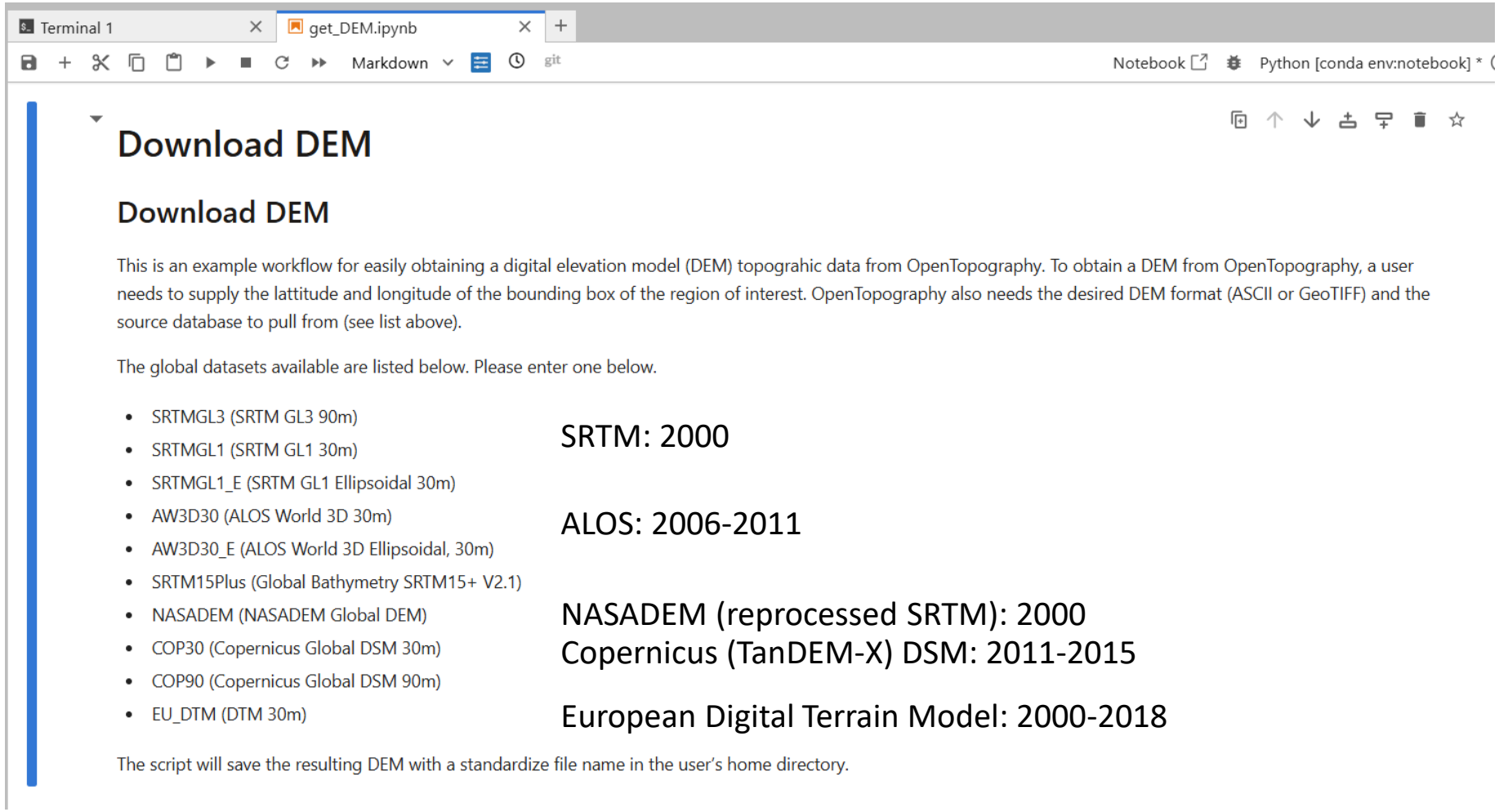
High-Resolution Topography Data and Tools

		ArcticDEM 2m
4	⊞ Global Ecosystem Dynamics Investigation (GEDI) L3 Gridded Land Surface Metrics Open Data	GEDI L3
5	⊞ Continental Europe Digital Terrain Model Open Data	Europe DTM
6	⊞ Copernicus Global Digital Elevation Models Open Data	Copernicus 30m Copernicus 90m
7	⊞ NASADEM Global Digital Elevation Model Open Data	NASADEM
8	⊞ Global Bathymetry and Topography at 15 Arc Sec: SRTM15+ V2.5.5 Open Data	SRTM15+
9	⊞ Global Multi-Resolution Topography (GMRT) Data Synthesis Open Data	GMRT
10	⊞ ALOS World 3D - 30m Open Data	AW3D30 AW3D30 Ellipsoidal
11	⊞ Shuttle Radar Topography Mission (SRTM) Global Open Data	SRTM GL3 SRTM GL1 SRTM GL1 Ellipsoidal

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Using VICTOR to get a DEM

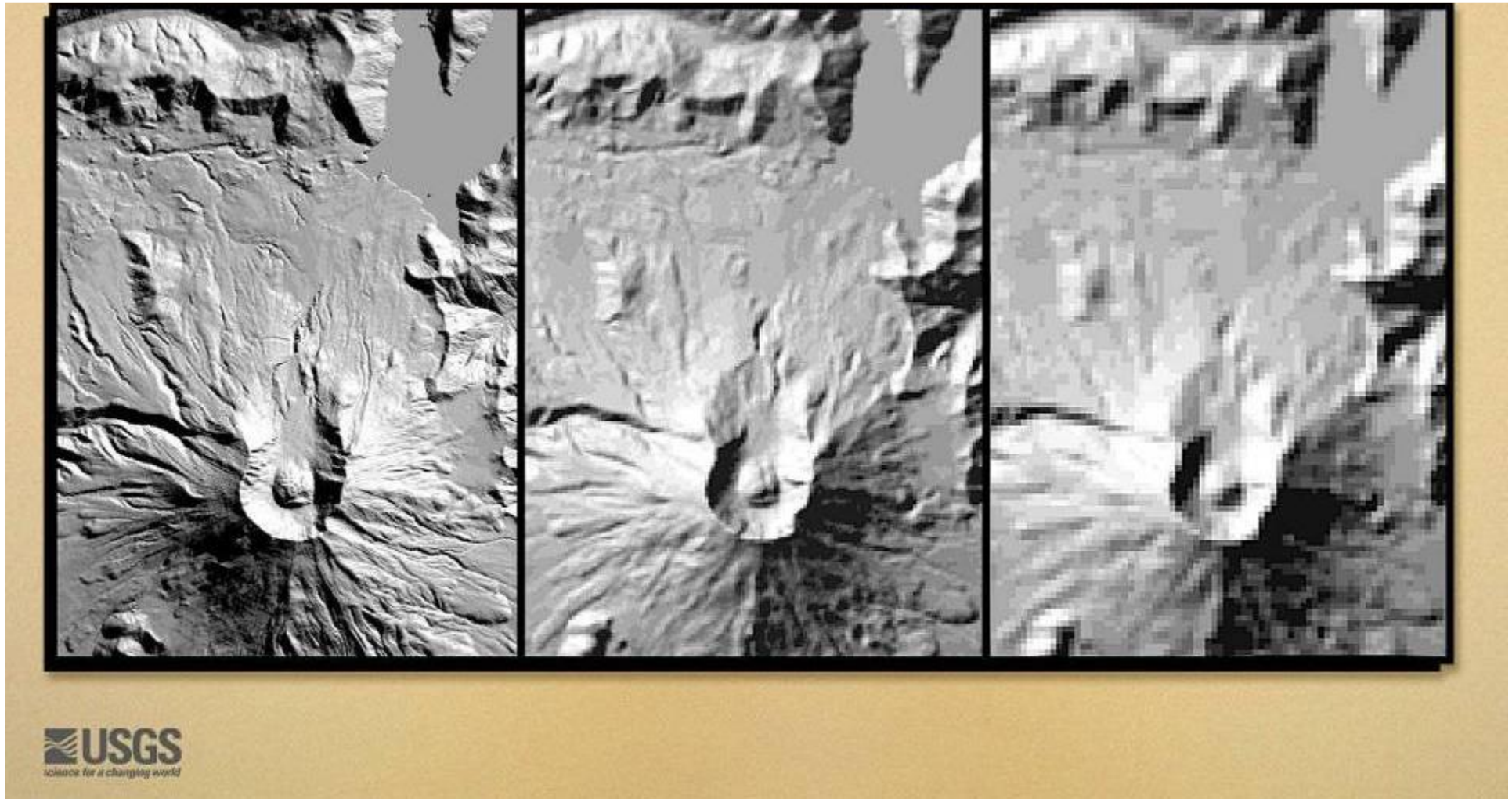


Which DEM would you want to use?

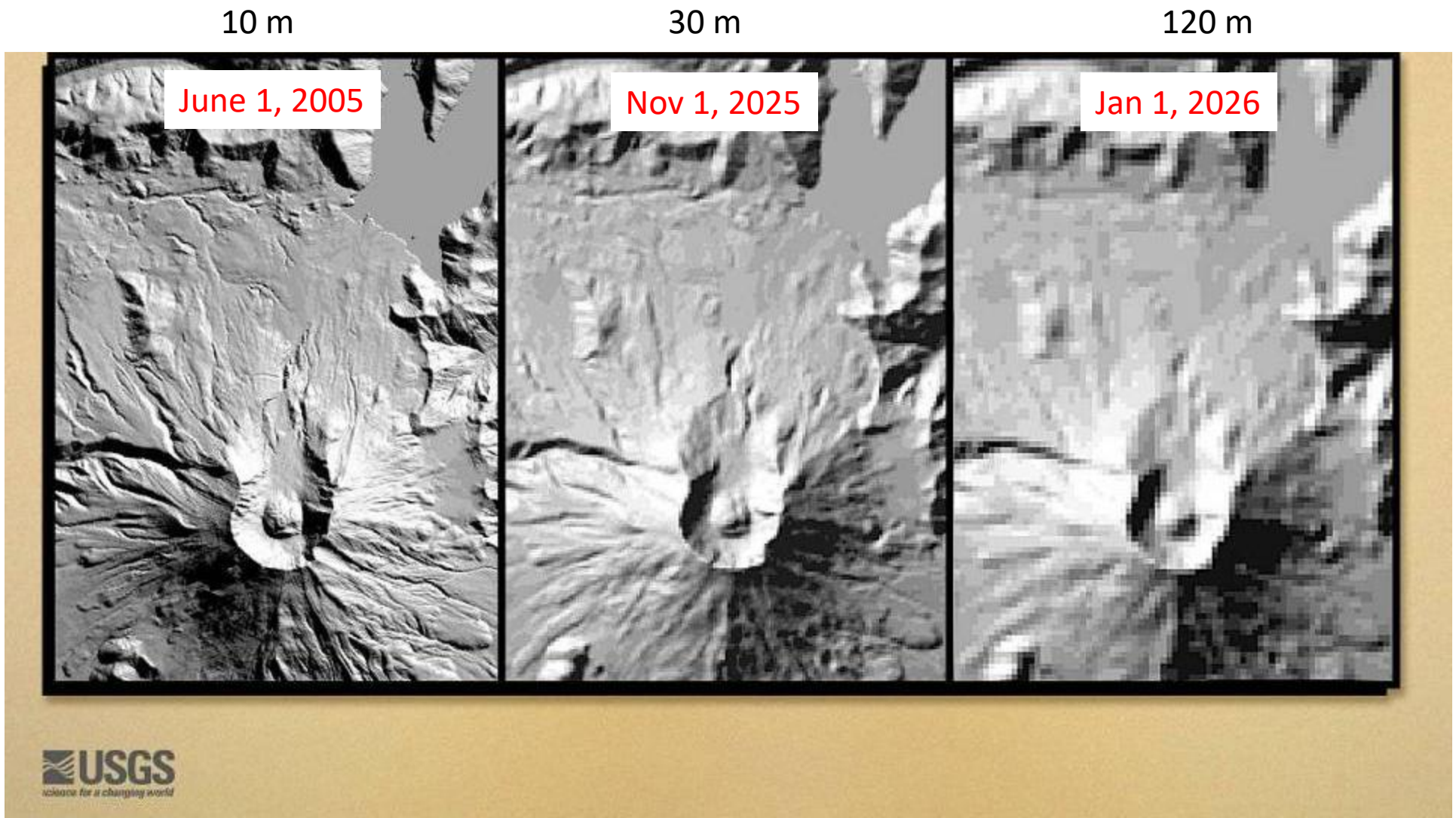
10 m

30 m

120 m



Which DEM would you want to use?



Which DEM would you want to use?

